

Corporation of the City of Quinte West

Quinte West Drinking Water System (Frankford/Batawa Service Area)

2024 Annual & Summary Drinking Water System Report





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Public Works and Environmental Services
Water/Wastewater Division
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Drinking Water System Information

Drinking Water System Number:	210001889
Drinking Water System Name:	Quinte West Drinking Water System (Frankford/Batawa Service area)
Drinking Water System Owner:	The Corporation of the City of Quinte West
Drinking Water System Category:	Large Municipal-Residential System
Period being Reported:	January 1, 2024 through December 31, 2024

Does your Drinking Water System service more than 10,000 people?

Yes.

Is your Annual Report available to the public at no charge on a website

Yes, please visit www.quintewest.ca.

Location where Summary Report required under O.Reg.170/03 Schedule 22 will be available for inspection

Water/Wastewater Services Administration Office

25 Couch Crescent

Trenton, ON, K8V 1G8

List all Drinking Water Systems (if any), that receive all of their Drinking Water from your System:

None.



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Indicate how you notified system users that your annual report is available, and is free of charge.

- ☒ Public access/notice via City website
- ☐ Public access/notice via Government Office
- ☐ Public access/notice via newspaper
- ☒ Public access/notice via City social media platform(s)
- ☐ Public access/notice via a Public Library
- ☐ Public access/notice via other method:

Description of the Drinking Water System

The Frankford Water Treatment Plant pulls water from the Trent River through a 240.5 m long, 450 mm diameter raw water intake pipe, extending 50 m into the Trent River approximately 240 m east of the Water Treatment plant. This conventional chemically assisted filtration plant has a rated capacity of 3,353 cu.m/day. The facility houses four (4) Ecodyne package plants. Processes used at the filtration plant include solids recirculating re-activator type flocculator/clarifier units with tube settlers and automated sludge withdrawal system, six (6) dual media high rate gravity filters, and two (2) Granular Activated Carbon (GAC) adsorption filters. Chlorine gas is applied as a disinfectant before filtered water enters two interconnected un-baffled clearwells with a combined capacity of 850 cu.m. The potable (drinkable) water is then pumped into the distribution system through a set of three high lift pumps that supply water to the distribution system, as well as the elevated water storage tower which has an operating capacity of 1,890 cu.m. A pressure reducing station located at 65 Parry Drive, provides system pressure protection within the service area of the Batawa community.

The Frankford/Batawa Drinking Water System (DWS) supplies water to approximately 3,900 people in the communities of Frankford and Batawa. In May of 2022 the City commissioned the Frankford Booster Station and a watermain along Glen Miller Road connecting the



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Trenton Water Distribution System (WDS) and Frankford/ Batawa WDS. The Frankford/ Batawa DWS previously governed by its own Drinking Water Works Permit (DWWP) and Municipal Drinking Water Licence (MDWL), was consolidated with the Trenton/ Bayside DWS, issued one DWWP, # 163-202, and one MDWL, # 163-102, the Quinte West DWS. Each system is still assigned a separate DWS number by the Ministry of the Environment, Conservation and Parks (MECP).

List all water treatment chemical used over this Reporting Period:

- ✓ Polyaluminum Chloride (Trade Name: SternPac)
- ✓ Chlorine Gas

Were any significant expenses incurred to:

- ✓ Install Required Equipment?
- ✓ Repair required Equipment?
- ✓ Replace required Equipment?

Provide a brief description, and a breakdown of monetary expenses incurred to facilitate Equipment upgrades:

Preventative Maintenance (PM) activities for lubrication, inspections, testing and cleaning of equipment is scheduled and completed routinely, along with other lifecycle replacement needs. In addition to the PM activity, the following Capital expenditures were incurred this Reporting Period:

Roof Replacement	~\$750,000
Intake Inspections	~\$152,000 (combined with all DWS)



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Water Meter Automation and Replacement	~\$120,000 (combined with all DWS)
Reactivated Carbon	~\$23,000
Building Security Upgrades	~\$14,000
GAC Pump Motor Replacement	\$7,500

Notices submitted in accordance with subsection 18(1) of the Safe Drinking Water Act or section 16-4 of Schedule 16 of O.Reg. 170/03 and reported to Spills Action Centre during this Reporting Period:

Incident Date (DD-MMM-YY)	Parameter	Result	Corrective Action	Corrective Action Date (DD-MMM-YY)
There were no notices submitted in accordance with subsection 18(1) of the SDWA or section 16-4 of Schedule 16 of O. Reg. 170/03 during this Reporting Period.				

Microbiological Testing completed in accordance with Schedule 10, 11, or 12 of Regulation 170/03 during this Reporting Period:

	Number of Samples	Range of E.Coli (min - max)	Range of Total Coliform (min - max)	Number of Heterotrophic Plate (HPC) Count Samples	Range of HPC results (min - max)
Raw	53	0 - NDOGT ¹	2 - NDOGT	NA	NA

¹ NDOGT stands for No Data: Overgrown with Target Bacteria. Prior, and subsequent datasets within normal range.



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Treated	53	0 - 0	0 - 0	53	0 - 1
Distribution	184	0 - 0	0 - 0	80	0 - 8

Operational Testing completed in accordance with Schedule 7, 8, or 9 of O.Reg.170/03 during this Reporting Period:

Parameter	Number of Grab Samples	Range of Results (min-max)
Turbidity, (NTU)	8760	0.015 - 1.291 ²
Primary Disinfection FAC ³ , (mg/L)	8760	1.37 ⁴ - 3.75 ⁵
Secondary Disinfection FAC, (mg/L)	8760	0.48 - 2.90

Summary of additional testing and sampling carried out in accordance with the requirement of an Approval, Order, or other Legal Instrument:

Not applicable.

² Isolated incident occurring for 5 seconds before filter effluent valve closed. Incident occurred on February 15, 2024.

³ Free Available Chlorine. Defined as the free amount of chlorine available in water.

⁴ Isolated incident suspected increased chlorine demand, operator responded accordingly. Incident occurred on March 18, 2024.

⁵ Isolated incident highlift pumps offline for period, operator batch dosing. Incident occurred on October 17, 2024.



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**Summary of Inorganic parameters tested during this Reporting Period
 or the most recent Sample Results:**

Inorganic Compound Results				
Parameter	Sample Date (dd/mmm/yy)	Result Value	Unit of Measure	Exceedance?
Antimony	03-JAN-24	< 0.6	ug/L	No
Arsenic	03-JAN-24	< 0.2	ug/L	No
Barium	03-JAN-24	29.1	ug/L	No
Boron	03-JAN-24	9	ug/L	No
Cadmium	03-JAN-24	< 0.003	ug/L	No
Chromium	03-JAN-24	0.13	ug/L	No
Mercury	03-JAN-24	< 0.01	ug/L	No
Selenium	03-JAN-24	< 0.04	ug/L	No
Sodium	03-JAN-24	10.2	mg/L	No
Fluoride	03-JAN-24	< 0.06	mg/L	No
Uranium	03-JAN-24	0.005	ug/L	No
Nitrate	03-JAN-24 03-APR-24 03-JUL-24 02-OCT-24	0.218 0.184 0.147 0.083	mg/L	No
Nitrite	03-JAN-24 03-APR-24 03-JUL-24 02-OCT-24	< 0.003 < 0.003 < 0.003 < 0.003	mg/L	No



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Summary of Lead Testing under Schedule 15.1 during this Reporting Period:

Lead Sampling Results				
Sampling Period	Location Type	Number of Samples	Range of Lead Results (ug/L), 'min-max'	Number of Exceedances
Winter Sampling Period ⁶	Hydrant	4	0.02 - 0.50	None
Summer Sampling Period ⁷	Hydrant	4	0.05 - 1.81	None

Summary of Organic parameters sampled during this Reporting Period:

Organic Compound Results				
Parameter	Sample Date	Result Value	Unit of Measure	Exceedance
Alachlor	03-JAN-24	< 0.02	ug/L	No
Atrazine + N-dealkylated metabolites	03-JAN-24	< 0.01	ug/L	No
Azinphos-methyl	03-JAN-24	< 0.05	ug/L	No

⁶ Winter Sampling Period runs from December 15 - April 15.

⁷ The Summer Sampling Period runs from June 15 - October 15.



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Benzene	03-JAN-24	< 0.32	ug/L	No
Benzo(a)pyrene	03-JAN-24	< 0.004	ug/L	No
Bromoxynil	03-JAN-24	< 0.33	ug/L	No
Carbaryl	03-JAN-24	< 0.05	ug/L	No
Carbofuran	03-JAN-24	< 0.01	ug/L	No
Carbon Tetrachloride	03-JAN-24	< 0.17	ug/L	No
Chlorpyrifos	03-JAN-24	< 0.02	ug/L	No
Diazinon	03-JAN-24	< 0.02	ug/L	No
Dicamba	03-JAN-24	< 0.20	ug/L	No
1,2-Dichlorobenzene	03-JAN-24	< 0.41	ug/L	No
1,4-Dichlorobenzene	03-JAN-24	< 0.36	ug/L	No
1,2-Dichloroethane	03-JAN-24	< 0.35	ug/L	No
1,1-Dichloroethylene (vinylidene chloride)	03-JAN-24	< 0.33	ug/L	No
Dichloromethane	03-JAN-24	< 0.35	ug/L	No
2-4 Dichlorophenol	03-JAN-24	< 0.15	ug/L	No
2,4-Dichlorophenoxy	03-JAN-24	< 0.19	ug/L	No



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acetic acid (2,4-D)				
Diclofop-methyl	03-JAN-24	< 0.40	ug/L	No
Dimethoate	03-JAN-24	< 0.06	ug/L	No
Diquat	03-JAN-24	< 1	ug/L	No
Diuron	03-JAN-24	< 0.03	ug/L	No
Glyphosate	03-JAN-24	< 1	ug/L	No
Malathion	03-JAN-24	< 0.02	ug/L	No
2 methyl-4-chlorophenoxyacetic acid (MCPA)	03-JAN-24	< 0.00012	mg/L	No
Metolachlor	03-JAN-24	< 0.01	ug/L	No
Metribuzin	03-JAN-24	< 0.02	ug/L	No
Monochlorobenzene	03-JAN-24	< 0.3	ug/L	No
Paraquat	03-JAN-24	< 1	ug/L	No
Pentachlorophenol	03-JAN-24	< 0.15	ug/L	No
Phorate	03-JAN-24	< 0.01	ug/L	No
Picloram	03-JAN-24	< 1	ug/L	No



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Polychlorinated Biphenyls (PCB)	03-JAN-24	< 0.04	ug/L	No
Prometryne	03-JAN-24	< 0.03	ug/L	No
Simazine	03-JAN-24	< 0.01	ug/L	No
Trihalomethanes (THMs) (Running annual average)	02-OCT-24	97.00	ug/L	No
Haloacetic acids (HAAs) (Running annual average)	02-OCT-24	63.28	ug/L	No
Terbufos	03-JAN-24	< 0.01	ug/L	No
Tetrachloroethylene	03-JAN-24	< 0.35	ug/L	No
2,3,4,6-Tetrachlorophenol	03-JAN-24	< 0.20	ug/L	No
Triallate	03-JAN-24	< 0.01	ug/L	No
Trichloroethylene	03-JAN-24	< 0.44	ug/L	No
2,4,6-Trichlorophenol	03-JAN-24	< 0.25	ug/L	No
Trifluralin	03-JAN-24	< 0.02	ug/L	No
Vinyl Chloride	03-JAN-24	< 0.17	ug/L	No



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Inorganic or Organic parameter(s) that have exceeded half the standard prescribed in Schedule 2 of Ontario Drinking Water Quality Standards:

None.



2024 Summary Report to Council

In accordance with Schedule 22 requirements outlined in Ontario Regulation 170/03, a Summary Report shall be prepared no later than March 31 for the preceding year, and supplied to members of municipal council.

The Report shall list the requirements of the Act, Regulations, Drinking Water Works Permit, Municipal Drinking Water Licence, and any Orders applicable to the system that were not met at any time during the period covered by the Report.

The Report must also include a summary of the quantities and flow rates of potable water (water intended for human consumption), supplied during the Reporting Period, including monthly average and maximum daily flows. A comparison of these flows to the rated capacity and flow rates approved in the system Drinking Water Works Permit, and Municipal Drinking Water Licence, must also be provided.

Prescribed Instruments applicable to the Frankford/Batawa DWS

The Frankford/Batawa Service Area is governed by, and must operate their DWS primarily in accordance with, the following Acts and Regulations at minimum:

- Safe Drinking Water Act, 2002;
- O. Reg. 128/04 – Certification of Drinking Water System Operators and WQA
- O. Reg. 170/03 – Drinking Water Systems
- O. Reg. 169/03 – Ontario Drinking Water Quality Standards
- Environmental Protection Act, where applicable;
- Clean Water Act, where applicable;
- Municipal Drinking Water Licence 163-102;
- Drinking Water Works Permit 163-202;
- Permit to Take Water Number P-300-9306939790, expires December 4, 2034



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Councillors wishing to obtain a copy of any Act or Regulation are free to contact the Compliance Coordinator at collette.kingdon@quintewest.ca to obtain a current consolidated copy and interpretation of the legislation.

Compliance with Prescribed Instruments, Acts and Regulations

Safe Drinking Water Act

Adverse Water Quality Incidents as defined by the Safe Drinking Water Act have been listed in section [Notices submitted in accordance with subsection 18\(1\) of the Safe Drinking Water Act or section 16-4 of Schedule 16 of O.Reg. 170/03 and reported to Spills Action Centre during this Reporting Period](#). The following noteworthy activities occurred throughout the reporting period.

NSF International conducted a Surveillance Audit on November 21 and 22, 2024. The intent of the audit is to confirm continued conformance to the Drinking Water Quality Management Standard. The City received its Re-Accreditation, with no Non-Conformances identified in the Audit Report. Two (2) Opportunities for Improvement (OFI) were identified.

The City achieved a 100% Inspection Rating on its 2024 MECP Compliance Inspection conducted on July 3, 2024.

Clean Water Act

The Source Protection Plan was approved by the Minister of Environment, Conservation and Parks, and came into effect on January 1, 2015. The City has put the necessary internal processes in place with the Planning, Building, and Public Works Departments to screen applications and ensure compliance with the Source Protection Plan. In 2024, thirty-one screening applications were submitted to the Risk Management Official for review and approval for the area surrounding the Frankford Intake Protection Zone. In general the



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screening, application, and approval processes implemented by staff seem to be working reasonably well. Throughout the screening and verification process it has been determined that there are five (5) Risk Management Plans (RMPs) required for affected properties in the Intake Protection Zone surrounding the Frankford Intake, which have all been finalized.

Permit to Take Water

The City operates its Frankford Water Treatment Plant in accordance with Permit to Take Water (PTTW) number P-300-9306939790 which expires on December 4, 2034. This Permit allows the City to take 3,045.8 cu.m/day from the Trent River watershed at a maximum flow rate of 62.5 L/s. The maximum recorded daily taking was 1,896 cu.m/day on June 18, 2024. The maximum recorded flow rate was 33.71 L/s on June 19, 2024.

For a detailed summary of water quantities and flow rates, see [2024 Water Quantities and Flow Rates - Raw Water Assessment](#).

Drinking Water Works Permit/Municipal Drinking Water Licence

The Frankford/Batawa DWS is now governed by Issue Number: 06 of Municipal Drinking Water Licence # 163-102, and Issue Number: 09 of Drinking Water Works Permit # 163-202. The Municipal Drinking Water Licence will expire September 28, 2025.

In accordance with Condition 1.1 of the MDWL, the Frankford Water Treatment Plant has a rated capacity to treat and distribute 3,353 cu.m/day. Over the Reporting Period the maximum daily volume of treated water distributed to the distribution system was 1,479 cu.m/day on June 18, 2024. The Annual Average Daily Flow pumped to the Distribution System was 1,043 cu.m/day. For a detailed summary of treated water quantities and flow rates, see [2024 Water Quantities and Flow Rates - Treated Water Assessment](#).



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2024 Water Quantity and Flow Rates (Raw Water Assessment):

Raw Water Flow: PTTW Limit = 3,045.8 cu.m/d			
Month	Average Daily Flow (cu.m/d)	Maximum Daily Flow (cu.m/d)	Total Monthly Flow (cu.m/month)
January	1472	1783	45632
February	1440	1614	41756
March	1333	1431	41334
April	1391	1446	41743
May	1590	1779	49284
June	1549	1896	46470
July	1555	1822	48195
August	1445	1710	44794
September	1365	1548	40943
October	1262	1445	39108
November	1307	1534	39195
December	1422	1519	44076
Total Raw Water Flow =			522,528



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2024 Water Quantity and Flow Rates (Treated Water Assessment):

Treated Water Flow: Facility Rated Capacity = 3,353 cu.m/day			
Month	Average Daily Flow (cu.m/d)	Maximum Daily Flow (cu.m/d)	Total Monthly Flow (cu.m/month)
January	1077	1414	33398
February	1016	1247	29467
March	937	1072	29053
April	983	1095	29489
May	1162	1397	36037
June	1164	1479	34933
July	1169	1390	36244
August	1078	1288	33409
September	1018	1168	30537
October	949	1085	29431
November	950	1052	28509
December	1010	1123	31302
Total Treated Water Flow =			381,810

- ➔ Annual Average Daily (Treated Water) Flow = 1,043 cu.m/day. This accounts for 31% of the facility Rated Capacity.
- ➔ Maximum Daily (Treated Water) Flow = 1,479 cu.m/day. This accounts for 44% of the facility Rated Capacity during recorded peak daily flow conditions.



Historical Flow Comparison

The 5-year ADF of Treated Water is calculated to be 1,389 cu.m/day, accounting for 41% of the facility rated capacity.

The decrease in ADF can be attributed to the installation of the Frankford Booster Station. The Trenton DWS provided 77,185 cu.m of treated water to the Frankford/ Batawa DWS in 2024.

